

SHANU KUMAR MANGLAM

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Portfolio: Shanu Kumar Manglam — GenAI Engineer

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Career Objective

Aspiring AI Engineer with hands-on experience in Generative AI, Retrieval-Augmented Generation (RAG), Machine Learning, and Cloud AI solutions. Skilled in developing AI-powered applications using Python, LLMs, Gemini, Vector Search, and cloud platforms, with experience building conversational agents, intelligent analytics solutions, and end-to-end AI workflows. Passionate about designing scalable, production-ready AI systems that solve real-world business challenges through innovation and automation.

Technical Skills

- **AI Agents & Frameworks:** Google ADK, LangChain, LangGraph, Agentic Workflows, Tool Calling, Multi-Agent Orchestration
- **Generative AI & RAG:** Retrieval-Augmented Generation (RAG), Vector Databases (FAISS), Embedding Models, Semantic Search, Prompt Engineering, LLMs (GPT-4o, Gemini API), Responsible AI
- **AI/ML & Computer Vision:** Machine Learning, Deep Learning, NLP, PyTorch, TensorFlow, Feature Engineering, Model Validation
- **Backend & Cloud DevOps:** Python, Flask, FastAPI, RESTful APIs, Streamlit, Docker, Google Cloud Run, GCP, AWS, Git, GitHub Actions, CI/CD, Playwright
- **Data Engineering & Analytics:** SQL, PySpark, ETL Pipelines, EDA, Data Cleansing, Data Quality, Power BI, Looker Studio
- **Methodologies & SDLC:** Agile, Scrum, Technical Documentation, Testing, Debugging, End-to-End AI Lifecycle

Projects

HealthBot | *Python, LangGraph, LangChain, GPT-4o Mini, Tavily API* Aug 2026 – Sep 2026

Github Link: github.com/Shanu1110/healthbot-patient-assistant

- Built a healthcare-focused RAG assistant leveraging LangGraph orchestration and LangChain pipelines to deliver verified medical information from trusted web sources.
- Engineered a multi-stage workflow integrating medical topic validation, web search, content summarization, quiz generation, and evaluation modules.
- Applied prompt engineering and structured LLM reasoning to automate knowledge assessment, tailored feedback, and continuous learning recommendations.

Coffee Barista RAG Agent | *Gemini 3.5 Flash, Vertex AI, Google ADK, Streamlit, Cloud Run* Jul 2026 – Aug 2026

Github Link: github.com/Shanu1110/coffee-barista-rag-agent

- Built a RAG-powered conversational AI assistant using Gemini 3.5 Flash, Vertex AI, and Google ADK for grounded, context-aware menu recommendations.
- Implemented allergen-aware filtering logic and custom retrieval pipelines to eliminate hallucinations and improve response reliability.
- Developed a Streamlit chat interface with persistent conversational memory and deployed the containerized service on Google Cloud Run using serverless architecture.

HR Policy Assistant – Enterprise RAG Application | *FAISS, LangChain, GPT-4o, FastAPI, Docker* May 2026

Github Link: github.com/Shanu1110/hr-policy-rag

- Engineered an enterprise RAG assistant using LangChain, GPT-4o, FAISS, and FastAPI for context-aware document question answering.
- Built semantic search pipelines (improving retrieval relevance by 35%), added safety guardrails, and containerized the service with Docker and CI/CD.

ATS Resume Copilot – Agentic Job Search Automation | *Gemini API, Playwright, LangChain, Streamlit* Apr 2026

Github Link: github.com/Shanu1110/ats-resume-copilot

- Designed an autonomous agent platform integrating Gemini API and Playwright to automate JD parsing, scoring, and web application workflows.
- Implemented session handling, browser automation tools, and prompt optimizations, cutting manual application effort by 50%.

Interactive Sudoku Web Engine | *Python, Flask, Backtracking Algorithm, REST APIs, HTML/CSS/JS* Jul 2026

Github Link: github.com/Shanu1110/sudoku-web-engine

- Architected a full-stack interactive Sudoku web application using Flask RESTful endpoints and an optimized backtracking solver algorithm.
- Implemented dynamic board generation, real-time validation APIs, state management, and difficulty-level customization.

Professional Experience

Cognizant <i>Programmer Analyst Trainee</i>	Chennai, India Feb 2026 – Apr 2026
- Engineered AI-driven agentic workflows, document summarization, and reporting using Microsoft Copilot Studio and prompt engineering with strict Responsible AI guardrails. - Built scalable ETL pipelines in Python and PySpark to cleanse, transform, and validate large structured marketing datasets for analytics and AI models. - Developed analytics dashboards integrating GA4, GTM, Adobe Analytics, and Looker Studio to track campaign KPIs in an Agile environment.	
Ernst & Young <i>Data Analyst Intern</i>	Remote Feb 2024 – Apr 2024
- Developed predictive analytics models and interactive Power BI KPI dashboards using Python and SQL to support healthcare data decisions. - Executed end-to-end EDA and ETL pipelines to cleanse and transform structured datasets, boosting reporting accuracy by 20%.	

Education

University of Engineering & Management, Kolkata <i>Bachelor of Technology in Electronics and Communication Engineering</i> (CGPA: 8.86 / 10)	Kolkata, India Aug 2021 – Jul 2025
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Awards & Honors

Vice Chancellor’s Award for Best Scientific Mind 2025 <i>Focus: Artificial Intelligence, Machine Learning, Electronics Engineering</i>	May 2025
Awarded for outstanding innovation and research contributions in AI, Machine Learning, and Electronics Engineering.	

Publications

Review Paper: Renewable Energy Trends for the Next Decade <i>UEMGREEN’25</i> — Published a review on AI-driven renewable energy technologies, grid modernization, and sustainable energy systems.	Apr 2025
Research Paper: Analysing CO2 Emission from Vehicles Using Machine Learning <i>UEMGREEN’25</i> — Developed and benchmarked ML models predicting vehicular CO2 emissions for sustainability analytics.	Apr 2025

Certifications

Generative AI & Agents: Google Cloud Gen AI Academy APAC 2026, Deploying RAG AI Agents with Google ADK + Cloud Run (Google Developer Program), ADK 2 Orchestration: Graph, Collaborative & Dynamic Workflows, Agentic AI Mastery, Generative AI, Prompt Engineering, Responsible AI Developer Tools & Cloud: Microsoft Copilot Studio, GitHub Copilot Complete Guide, Apache Spark 3, AWS Cloud Fundamentals, Google Analytics (GA4)	
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